12-week Plan
Title: Predictive Pre-Congestion Aware QoS Optimization for Video Streaming using SDN and Lightweight AI

Team:

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Problem Statement – Key Points

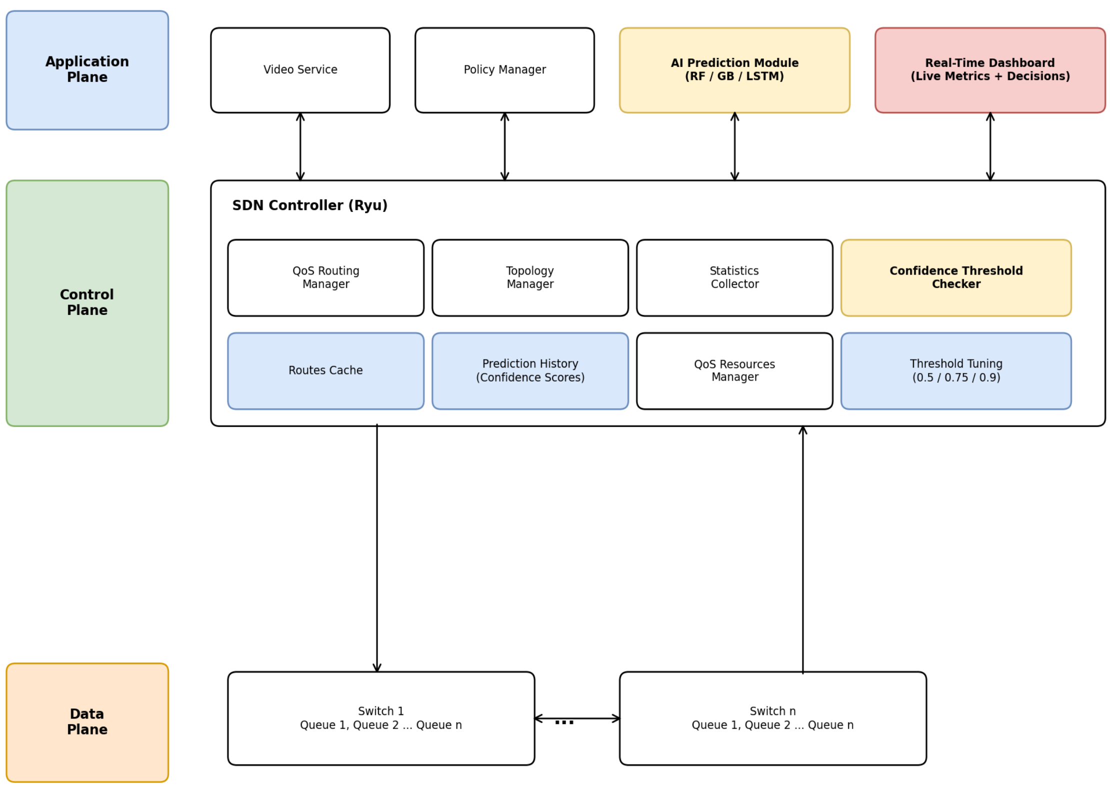
- Video streaming applications suffer from buffering, quality degradation, and increased latency when network congestion occurs, directly impacting Quality of Experience (QoE).
- Traditional QoS mechanisms in SDN, such as static priority queuing and fixed routing rules, are reactive — they respond only after congestion is detected, by which point video quality has already degraded.
- Existing SDN-based QoS solutions for video streaming (e.g., adaptive routing approaches such as ARVS, flow maximization frameworks such as MaxStream) primarily optimize routing after congestion is observed, and largely rely on bandwidth/throughput as the sole optimization metric.
- There is limited work combining lightweight, real-time predictive models with SDN controllers to anticipate congestion before it occurs, specifically for video traffic prioritization.
- Most AI-assisted video streaming research focuses on client-side bitrate adaptation, not network-side proactive control at the SDN controller level.
- A gap exists for a lightweight, deployable prediction model that can output a confidence score and run within real-time constraints of an SDN controller, avoiding unnecessary rerouting triggered by noisy or low-confidence predictions.

Key Features of Proposed System

- Proactive, prediction-based congestion handling (not reactive) — the SDN controller acts before congestion occurs, not after.
- Confidence-aware triggering — rerouting/prioritization only occurs when the model's prediction confidence exceeds a tuned threshold, reducing false positives.
- Multi-model benchmarking — multiple ML models (Random Forest, Gradient Boosting, LSTM) are trained and compared, with the best-performing model selected based on measured results rather than assumption.
- Real-time visualization dashboard — live network metrics and AI-driven decisions are displayed during demonstration, providing transparency into system behavior.

Architecture Diagram

Architecture Diagram — Predictive Pre-Congestion Aware QoS Optimization for Video Streaming using SDN



Proposed Solution:

Month 1: Foundation and Setup

- Week 1–2: Literature Review
  - Study SDN architecture: OpenFlow protocol, controller–switch communication, flow tables.
  - Review existing reactive SDN-QoS approaches (e.g., ARVS, MaxStream) and identify their limitations.
  - Study lightweight time-series prediction techniques (e.g., moving average baselines, gradient boosting, simple LSTM) suitable for real-time congestion forecasting.
- Week 3: Tool Familiarization
  - Set up testbed using Mininet.
  - Install and configure Ryu SDN controller; verify controller–switch communication.
  - Simulate basic video-like traffic flows using iperf3.
- Week 4: Baseline Experiments
  - Build a multi-switch, multi-host topology in Mininet.
  - Run baseline experiments using standard reactive/static QoS policy (no AI).

- Measure: Latency, Jitter, Packet loss, Throughput under increasing congestion.

Month 2: Implementation and Optimization

- Week 5–6: Data Collection & Feature Engineering
    - Generate varied traffic load scenarios (light, moderate, bursty, congested) using iperf3.
    - Log time-series network statistics (queue length, bandwidth utilization, jitter trend, flow arrival rate).
    - Construct windowed features for short-horizon forecasting (predict congestion 2–5 seconds ahead).
- Week 7: Predictive Model Development
    - Train and benchmark multiple lightweight prediction models — e.g., Random Forest, Gradient Boosting, and a small LSTM — to forecast near-future congestion likelihood, with an associated confidence score for each prediction.
    - Compare models on accuracy, precision/recall, and prediction latency; select the best-performing model for integration.
    - Benchmark prediction latency to confirm real-time feasibility.
    - Validate accuracy on a held-out test set.
- Week 8: Testing Enhanced Setup (AI–SDN Integration)
    - Integrate the trained model into the Ryu controller's decision loop.
    - Implement proactive flow reprioritization/rerouting, triggered only when predicted congestion confidence exceeds a tuned threshold (e.g., test thresholds such as 0.5, 0.75, 0.9).
    - Run comparative experiments with: Proactive AI-driven QoS (optimized), Reactive/static QoS (baseline).
    - Collect performance metrics across multiple flows and varying conditions (e.g., bursty traffic, multiple congestion events).

Month 3: Analysis, Documentation, and Presentation

- Week 9–10: Data Analysis
    - Plot latency, jitter, packet loss, throughput, and a derived QoE score.
    - Compare: Reactive/static QoS vs. Proactive AI-driven QoS.
    - Present comparative benchmark results across the multiple ML models trained in Week 7 (accuracy, latency, false-positive rate) to justify the final model choice.
    - Evaluate the effect of different confidence thresholds on false-positive rerouting versus missed/delayed reactions; identify the optimal threshold.
    - Analyze the prediction lead-time benefit — how much earlier congestion is mitigated compared to reactive approaches.
    - Build a real-time dashboard (using Python/Matplotlib animation or a lightweight web interface) to visualize live network metrics and AI-driven decisions during demonstration.
- Week 11: Report and PPT Preparation
    - Draft project report (IEEE conference format) on Overleaf.
    - Prepare slides: Problem, Motivation, Architecture, Results, Future Work.
    - Create visual diagrams (problem statement, solution, architecture, graphs) using Draw.io.

- Week 12: Final Review and Submission
  - Conduct an internal mock presentation.
  - Submit code/test scripts on GitHub.
  - Submit final report + presentation deck to faculty.

Tools & Technologies to be used

- Mininet (network emulation)
- Ryu Controller (SDN controller, Python-based)
- iperf3 (traffic generation)
- Wireshark (packet inspection)
- Python (scikit-learn / lightweight LSTM via TensorFlow-Keras)
- Matplotlib / Plotly / lightweight web interface (real-time dashboard for visualization)
- Matplotlib / Excel for plotting

References: (6-10 references)

1. L. Ai, H. Zhang, and Y. Wang, "Adaptive routing for video streaming with QoS support over SDN networks," IEEE Conference Publication.
2. S. Kaur, K. Kumar, and N. Singh, "MaxStream: SDN-based Flow Maximization for Video Streaming with QoS Enhancement," IEEE Conference Publication.
3. Q. Zhang et al., "QoE-Driven Adaptive Video Streaming: Architectures, Techniques, and Future Research Challenges Toward 6G Networks," IEEE Journals & Magazine.
4. R. Kumar et al., "Optimizing QoE in IoT-Based Video Streaming through Deep Learning Algorithms," IEEE Conference Publication.
5. R. Farahani et al., "Towards AI-Assisted Sustainable Adaptive Video Streaming Systems: Tutorial and Survey."
6. Adaptive Video Streaming with AI-Based Optimization for Dynamic Network Conditions.

Additional points:

Literature Survey and Reference Documents:

- Preferably the last 3 years - IEEE Transactions such as: Mobile Computing, Networks and Service Management, Cognitive Communications and Networking, Wireless Communications, Networking, etc.
- IEEE Communications Surveys and Tutorials (Relevant topics or combinations of topics: Survey Papers – preferably in last 3 years).
- IEEE flagship conferences such as: INFOCOM, ICC, GLOBECOM, etc.
- ACM flagship conferences such as: SIGCOMM, MobiCom, CoNEXT, etc.

Standards: IEEE, IETF, ITU, 3GPP (latest releases), NIST, and related